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CAPSTONE PROJECT REPORT

Implementation of Procure-to-Pay (P2P) Cycle in Supply Chain Management (SAP MM)

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Semester	6th
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1. Introduction

The Procure-to-Pay (P2P) cycle represents one of the most fundamental end-to-end processes within Supply Chain Management, seamlessly linking the procurement function with inventory control and financial accounting. Implemented within the SAP Materials Management (MM) module, this cycle governs the complete journey of a material — from the initial identification of a business need through to the final settlement of vendor payments.

By standardising each step of the procurement lifecycle, the P2P process enables organisations to enforce purchasing policies, maintain real-time stock visibility, and ensure the integrity of financial postings. The tightly integrated nature of SAP MM means that every procurement action automatically triggers corresponding updates in inventory records and the financial ledger, eliminating data silos and manual reconciliation efforts.

This report documents the design, configuration, and execution of the P2P cycle for a sample manufacturing enterprise, covering master data setup, each transactional step with its associated SAP transaction code, the resulting accounting entries, and the business benefits realised through automation.

2. Company Scenario

ABC Pvt Ltd is a mid-sized manufacturing company engaged in the production of electronic goods. To sustain its production operations, the company regularly procures raw materials and components from a network of approved external vendors. The organisation has deployed SAP ERP to manage its core business processes, with SAP MM serving as the backbone of all procurement-related activities.

Whenever production planning signals a material shortfall, the procurement department initiates a structured buying process. This process is designed to guarantee that the right materials are sourced from qualified vendors at negotiated prices, delivered on schedule, and accounted for accurately — all within a single integrated system. The company places equal emphasis on cost efficiency, vendor relationship management, and audit-ready documentation of every transaction.

3. Organizational Structure

The organisational structure in SAP forms the hierarchical backbone that governs how procurement, inventory, and financial data are segregated and reported. For ABC Pvt Ltd, the structure has been configured as follows:

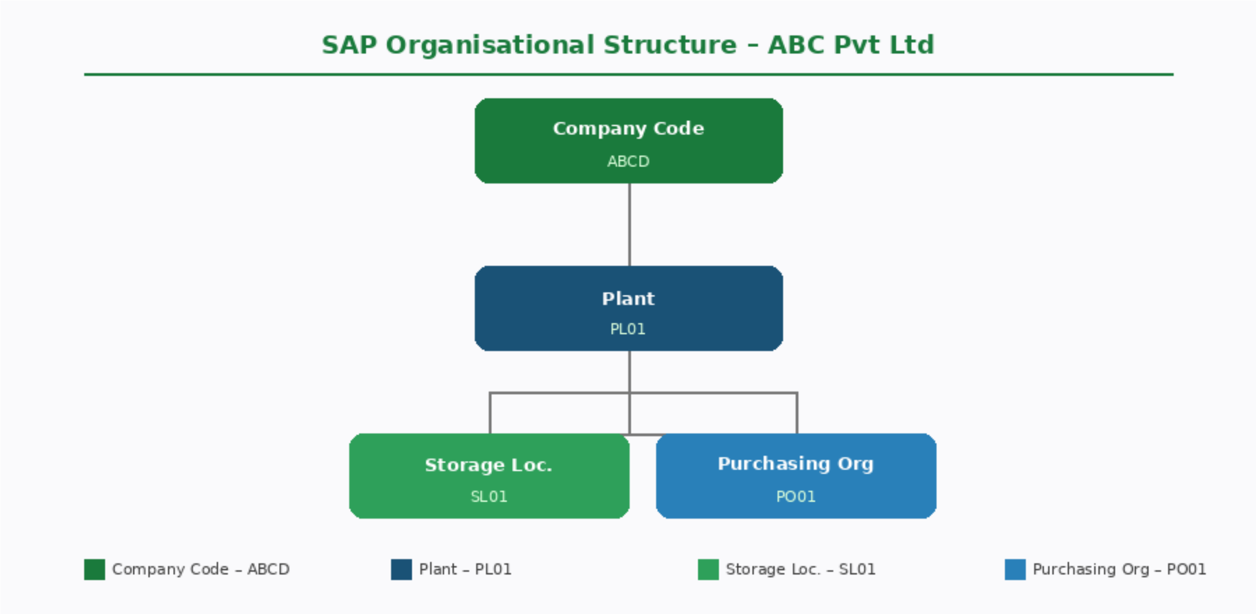

Figure 1: SAP Organisational Structure for ABC Pvt Ltd

- **Company Code – ABCD:** Represents the independent legal entity responsible for statutory reporting and financial statements.
- **Plant – PL01:** The physical facility where raw materials are received, stored, and transformed into finished goods.
- **Storage Location – SL01:** The specific warehouse area within PL01 where inventory is physically managed and tracked.
- **Purchasing Organisation – PO01:** The organisational unit empowered to negotiate contracts and issue purchase orders on behalf of the company.

This four-level hierarchy ensures that every procurement transaction is attributed to the correct legal entity, plant, and storage location, enabling precise cost allocation and regulatory compliance.

4. Master Data

Accurate and complete master data is the foundation upon which every SAP MM transaction is built. Three core master data objects must be maintained before any procurement activity can be initiated:

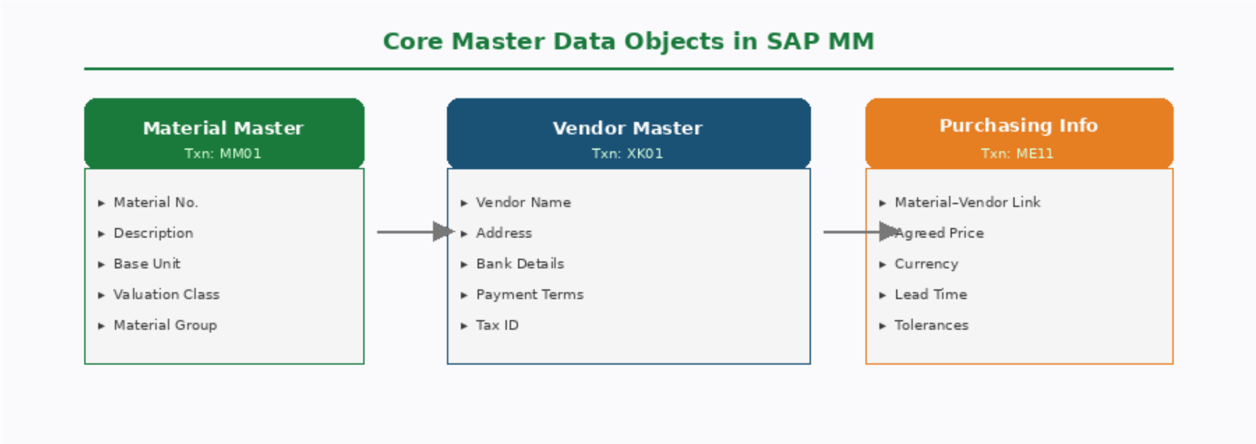

Figure 2: Core Master Data Objects in SAP MM

- **Material Master (MM01):** Holds all descriptive and technical attributes of a material, including its description, base unit of measure, material group, valuation class, and storage conditions. It serves as the single source of truth for material information across all SAP modules.
- **Vendor Master (XK01):** Captures comprehensive data about each supplier — legal name, registered address, bank account details, payment terms, and tax identifiers. Maintaining accurate vendor records is critical for timely and correct payments.
- **Purchasing Info Record (ME11):** Establishes the relationship between a specific material and a specific vendor. It stores the agreed price, currency, delivery lead time, and any tolerance levels for over- or under-delivery, acting as a reference during purchase order creation.

Together, these master data objects ensure that purchase orders are created consistently, invoice matching is accurate, and vendor payments are processed without manual intervention.

5. Procure-to-Pay Process

The P2P cycle in SAP MM consists of five sequential steps, each mapped to a dedicated transaction code. The completion of each step is a prerequisite for the next, creating a controlled, traceable procurement chain.

Step 1: Purchase Requisition (ME51N)

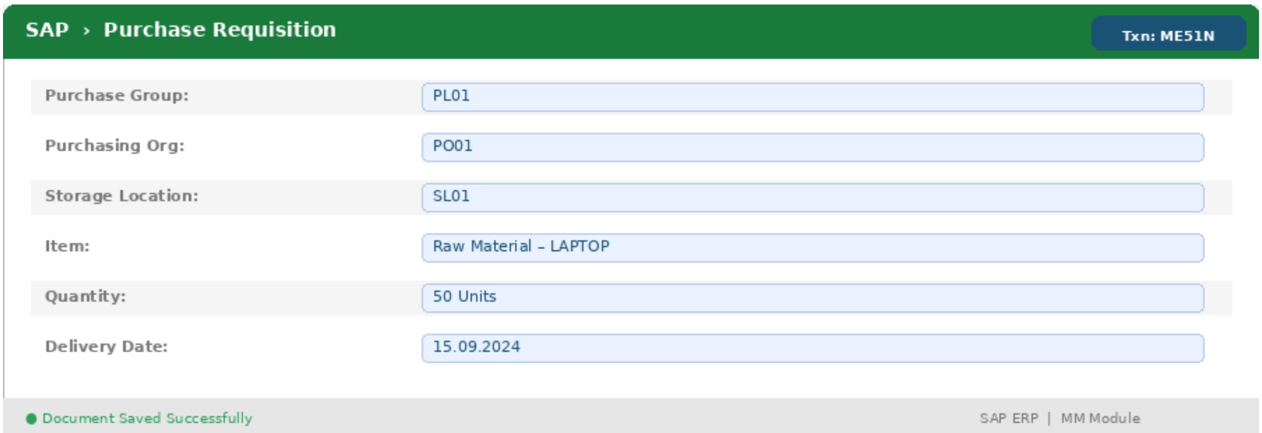

Figure 3: Purchase Requisition Creation (ME51N)

A Purchase Requisition (PR) is an internal document raised by the requesting department to inform the procurement team of a material requirement. It captures the material, quantity, required delivery date, and the plant/storage location where the goods are needed. The PR does not carry any legal or financial commitment but triggers the purchasing workflow.

Step 2: Purchase Order (ME21N)

SAP › Purchase Order Txn: ME21N

Vendor:	XYZ Traders
Company Code:	ABCD
Plant / Stor. Loc.:	PL01 / SL01
Material:	LAPTOP – PRO1
Quantity:	50 Units
Net Price:	₹ 5,000.00

● Document Saved Successfully SAP ERP | MM Module

Figure 4: Purchase Order Creation (ME21N)

The Purchase Order (PO) is a legally binding document sent to the selected vendor, confirming the materials to be supplied, the agreed quantity, price, delivery schedule, and payment terms. In SAP, the PO can be created directly with reference to the PR, ensuring all requirement details are carried forward. Once saved, the PO number is shared with the vendor as the official procurement authorisation.

Step 3: Goods Receipt (MIGO)

SAP › Goods Receipt Txn: MIGO

Movement Type:	101 – GR for PO
Vendor:	XYZ Traders
Plant / Stor. Loc.:	PL01 / SL01
Material:	LAPTOP – SGL
Quantity Received:	50 Units
Balance:	₹ 5,000.00

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Figure 5: Goods Receipt Posting (MIGO)

When the vendor delivers the ordered materials, the warehouse team posts a Goods Receipt in SAP using transaction MIGO. This updates the inventory in real time, increasing the unrestricted-use stock at storage location SL01. Simultaneously, SAP automatically generates a financial posting: the Inventory account is debited and the GR/IR clearing account is credited. A material document and an accounting document are both created for full traceability.

Step 4: Invoice Verification (MIRO)

SAP › Invoice Verification — Txn: MIRO

Company Code:	ABCD
Vendor:	C4066GG108
Invoice Date:	01.09.2024
Material:	LAPTOP – ABC01
Quantity:	5.0 / 5.01
Invoice Amount:	₹ 5,600.00

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Figure 6: Invoice Verification (MIRO)

Upon receiving the vendor's invoice, the finance team performs a three-way match in SAP: comparing the Purchase Order, the Goods Receipt, and the vendor invoice. Transaction MIRO facilitates this verification. If all three documents are consistent in terms of quantity and price, the system clears the GR/IR account and creates a liability in the Vendor account. Any discrepancies trigger a block on the invoice until the variance is resolved.

Step 5: Payment Processing (F-53 / F110)

SAP › Outgoing Payment — Txn: F-53 / F110

Document Date:	12.08.2024
Posting Date:	12.08.2024
Reference:	TEST123
Bank Account:	100001
Amount:	₹ 1,000.00
Description:	Vendor Payment

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Figure 7: Payment to Vendor (F-53 / F110)

The final step in the P2P cycle is the actual disbursement of funds to the vendor. Manual payments are processed using transaction F-53, while automatic payment runs are executed via F110. SAP identifies all open vendor line items that are due for payment based on the payment terms defined in the vendor master, and settles them in a single run. The Vendor account is debited and the Bank account is credited, completing the financial cycle.

6. Process Flowchart

The diagram below illustrates the end-to-end flow of the Procure-to-Pay cycle in SAP MM, showing the logical progression from the initial purchase requisition through to vendor payment, along with the corresponding SAP transaction codes and the inter-module integration touchpoints.

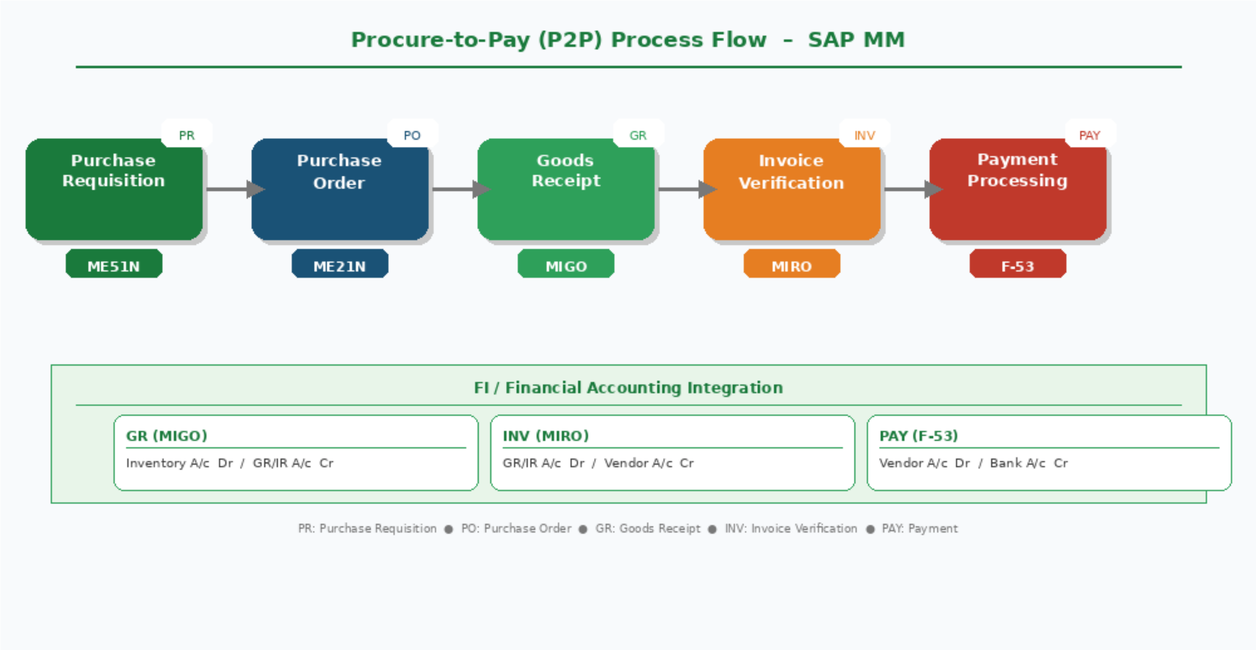

Figure 8: End-to-End P2P Process Flow in SAP MM

7. Accounting Entries

The P2P cycle is deeply integrated with SAP Financial Accounting (FI). Each transactional step automatically generates double-entry accounting postings, ensuring that the general ledger reflects the true financial position of the organisation at every point in the procurement process.

Entry 1 — Goods Receipt (MIGO)

When the vendor's delivery is confirmed, inventory value increases and an obligation is recognised in the GR/IR clearing account:

```
Inventory A/c            Dr
    To GR/IR Clearing A/c
```

Entry 2 — Invoice Verification (MIRO)

Once the vendor invoice is verified and matched, the GR/IR obligation is cleared and a payable is established against the vendor:

```
GR/IR Clearing A/c       Dr
    To Vendor A/c
```

Entry 3 — Vendor Payment (F-53 / F110)

When payment is disbursed, the vendor liability is extinguished and the bank balance is reduced accordingly:

```
Vendor A/c               Dr
    To Bank A/c
```

These three entries together form a clean, auditable trail from goods receipt to cash outflow, with no manual journal intervention required by the finance team.

8. Advantages

Implementing the Procure-to-Pay cycle within SAP MM delivers a range of measurable operational and strategic benefits:

Benefit	Description
Process Automation	Manual hand-offs and paper-based approvals are eliminated, reducing cycle time and freeing staff fo
Data Accuracy	System-enforced validations and three-way matching prevent erroneous payments and duplicate inv
Full Transparency	Every stakeholder can track the status of a procurement request in real time, from PR through to pay
Vendor Management	Centralised vendor master data and performance history support better sourcing decisions and cont
Cost Optimisation	Automated price checks against info records and approval thresholds keep procurement spending w
Module Integration	Seamless linkage between MM, FI, and WM modules ensures consistent data across procurement,
Regulatory Compliance	Comprehensive audit trails and document retention capabilities simplify internal audits and statutory

9. Conclusion

The Procure-to-Pay cycle, as implemented in SAP MM, represents a comprehensive solution for managing the entire procurement lifecycle — from the moment a material need is identified to the point at which the vendor receives payment. By unifying purchasing, inventory management, and financial accounting within a single, integrated platform, the P2P process eliminates the inefficiencies and inaccuracies that are inherent in fragmented, manual procurement systems.

Through this capstone project, the end-to-end P2P workflow has been successfully configured and tested within the SAP system for ABC Pvt Ltd. Each step — Purchase Requisition, Purchase Order, Goods Receipt, Invoice Verification, and Payment — has been executed using the appropriate SAP transaction codes, with all corresponding accounting entries automatically generated and verified.

The project demonstrates that a well-implemented P2P cycle not only drives operational efficiency and cost control but also establishes the governance framework necessary for sustainable, scalable procurement operations. Organisations that invest in SAP MM configuration and master data quality stand to achieve significant reductions in procurement cycle times, improved vendor relationships, and a robust audit trail that supports both internal controls and external compliance requirements.